

MCP Servers & Custom Tool Integration

Agentic Workflows | Advanced

Agenda

Time	Topic
0:00–0:16	MCP model, discovery, and transports
0:16–0:26	Local configuration
0:26–0:34	Approved local demo
0:34–0:48	Custom server walkthrough
0:48–1:00	Safety review and lab handoff

MCP model

Part	Meaning
Client	AI surface that requests work
Server	Process or service exposing capabilities
Transport	stdio, SSE, or Streamable HTTP
Tool	Callable capability
Resource	Data available from the server

The client discovers tools, calls one, and returns the result to the model.

MCP is a protocol boundary

MCP gives a model structured ways to discover and call capabilities. It does not make a capability safe, accurate, or approved.

The server owner still decides:

- which tools and resources exist;
- which inputs are accepted;
- what identity and permissions each call uses;
- what the server logs and returns on failure.

Treat each tool as an API surface with an agent in front of it.

Tool descriptions are part of the interface

Models use names, descriptions, and schemas to decide whether and how to call a tool. Vague descriptions create vague requests.

Weak	Better
<code>search</code>	<code>search_team_members</code>
"Find data"	"Search synthetic team records by name, role, or team"
<code>query: string</code>	<code>query</code> with format, limits, and empty-result behavior

Return data the model can cite and a clear error for an invalid request. Do not hide a failure behind an empty success-shaped response.

Transport changes the risk profile

Transport	Typical boundary to review
stdio	Local command, package source, files, and process permissions
SSE	Network endpoint, authentication, and session handling
Streamable HTTP	Endpoint, identity, authorization, and request limits

Start with the smallest approved path. A local server can still read sensitive files or run unsafe commands, and a remote server can widen the data boundary.

Local configuration

```
{
  "servers": {
    "sqlite": {
      "command": "uvx",
      "args": ["mcp-server-sqlite", "--db-path", "./data/app.db"]
    }
  }
}
```

Use `.vscode/mcp.json` for repository-shared configuration. Verify current setting names and supported behavior in official documentation.

A custom typed tool

```
server.tool(  
  "search_team",  
  "Search team members by name, role, or team. Returns matching results.",  
  { query: z.string().describe("Name, role, or team to search") },  
  async ({ query }) => ({ content: [{ type: "text", text: query }] })  
);
```

Clear descriptions and typed inputs help the model call the tool correctly.

Safety boundary

- Use supplied synthetic data only.
- Inspect the server source, tool descriptions, inputs, and outputs.
- Stop at the customer-defined metered-work guard.
- Do not connect credentials or customer systems.
- Session 17 covers organization policy, approval, registry, and rollout.

Lab

Build a weather server with `get_weather`, `get_forecast`, and `convert_temperature`. Test it locally, then use the supplied scenarios.